Statistical analysis plan (SAP): FLUID-ICU

> **SIMULATED STUDY: FOR TEACHING.** FLUID-ICU and its data are invented. This SAP shows what a plan written **before** the analysis looks like. It follows the 14 sections of `Course/Templates/Statistical_Analysis_Plan_Template.docx`.

Study title	FLUID-ICU: cumulative fluid balance in the first three ICU days and outcomes in adults with sepsis. A multicentre prospective cohort study
Protocol version	2.0 (simulated)
SAP version / date	1.0, signed before database lock
Written by	Study statistician and principal investigator (course team)

1. Study summary and objectives

Multicentre prospective cohort study. Consecutive adults (18 years or older) admitted with sepsis (Sepsis-3) to 4 ICUs (ICU A-D) over 2 years (January 2023 to December 2024). Patients receive usual care; the study only records data.

- **Primary objective:** to estimate the association between a positive cumulative fluid balance at day 3 and 28-day all-cause mortality, adjusted for pre-specified confounders.
- **Secondary objectives:**
  1. association between cumulative fluid balance (ml/kg) and acute kidney injury (AKI);
  2. association between cumulative fluid balance and ventilator-free days to day 28;
  3. association between admission lactate and 28-day mortality, and the effect of missing lactate;
  4. association between positive balance and ICU length of stay among 28-day survivors.

Each secondary objective matches one of the course questions (Q2-Q5); the primary objective is Q1.

2. Outcomes and their definitions

Outcome	Type	Definition (variables)	Time point
Primary: all-cause mortality	Binary	`died_28d` = Yes	Day 28
Time to death	Time-to-event	`time_to_event_28d` (days from ICU admission to death; survivors censored at 28) with `died_28d`	Up to day 28
AKI	Binary (ordinal stage)	`aki_any` = KDIGO stage >= 1 (`aki_stage`, maximum stage, ICU days 1-7)	Days 1-7
Ventilator-free days	Count 0-28	`vent_free_days_28`; 0 if died before day 28	Day 28
ICU length of stay	Continuous, skewed	`icu_los_days`, analysed in 28-day survivors	ICU discharge

Exposure. Cumulative fluid balance at day 3, `cum_fluid_balance_d3_ml_kg` = (fluid in days 1-3 - fluid out days 1-3) / admission weight (ml/kg). Analysed as binary `positive_balance_d3` (> 0 ml/kg) for the primary objective and as continuous (per 10 ml/kg) for AKI and ventilator-free days.

3. Analysis populations

Population	Definition	Used for
Enrolled	All patients after removal of duplicate records	Description, Table 1, missing-data summary, Q4
Exposure population	Enrolled patients with a computable day-3 fluid balance	Q1, Q2, Q3
Complete-case population	Exposure population with no missing value in the model variables	Adjusted models (main for Q1-Q3, Q5; sensitivity for Q4)
Survivors	Exposure population alive at day 28	Q5

A STROBE flow diagram will report the number at each step.

4. Descriptive analysis (Table 1)

Baseline characteristics by fluid-balance group (negative vs positive), with an overall column. Continuous variables: mean (SD) if roughly symmetric (age, weight), otherwise median (IQR) (SOFA, APACHE II, lactate, MAP). Categorical variables: n (%). Number missing reported for each variable. **No p-values in Table 1**: it describes the groups; it does not test them.

5. Primary analysis

Outcome	28-day mortality, exposure population
Method	Multivariable logistic regression
Effect measure	Odds ratio (positive vs negative balance) with 95% CI; the crude OR and the risk difference are also reported
Adjustment	Age, SOFA score, lactate, vasopressor, mechanical ventilation, chronic kidney disease (see section 7)
Significance	Two-sided, 5% level. The estimate and its CI are the result; the p-value supports it

Supportive analysis: Kaplan-Meier curves by group with numbers at risk and the log-rank test; Cox proportional-hazards model (crude and adjusted for the same confounders), hazard ratio with 95% CI; proportional hazards checked with Schoenfeld residuals (`cox.zph`).

6. Secondary analyses (supportive / exploratory)

Objective	Summary	Test / model	Effect measure
AKI (Q2)	n (%) by group and by balance category (<= 0, > 0-20, > 20-40, > 40 ml/kg)	Chi-square; logistic regression, balance per 10 ml/kg, adjusted as in section 7	OR per 10 ml/kg (95% CI)

Objective	Summary	Test / model	Effect measure
Ventilator-free days (Q3)	Median (IQR) by group; distribution shown	Mann-Whitney; linear regression, balance per 10 ml/kg, adjusted	Difference in days (95% CI); Hodges-Lehmann difference
Lactate (Q4)	Median (IQR) by vital status; mortality by lactate category (< 2, 2-4, > 4 mmol/L)	Logistic regression, lactate per 1 mmol/L, adjusted for age, sex, diabetes, CKD, heart failure; second model adds SOFA	OR per 1 mmol/L (95% CI)
ICU stay (Q5)	Median (IQR), survivors	Mann-Whitney; linear regression on log(ICU stay), adjusted	Ratio of geometric means (95% CI)

Group comparisons that are not in the table (age, SOFA, lactate between groups) use the t-test for symmetric variables and the Mann-Whitney test for skewed variables or scores.

7. Confounders and adjustment

Chosen **in advance** from clinical knowledge with a simple causal diagram: sicker patients receive more fluid and die more often (**confounding by indication**). Confounders: age; severity (SOFA score, lactate, vasopressor, mechanical ventilation on admission); chronic kidney disease (affects fluid prescribing and outcomes). APACHE II overlaps with SOFA and age and is not added. Variables measured after day 3 (AKI, RRT) are outcomes or mediators and are **not** adjusted for. With about 160-180 deaths and 7 model terms there are more than 20 events per term.

8. Subgroups and interaction (pre-specified)

One pre-specified effect modifier: **heart failure** (expected direction: a stronger association between positive balance and death in heart failure, because these patients tolerate fluid less well). Tested with a positive_balance_d3 x heart_failure interaction term in the primary model (Wald and likelihood-ratio tests). The ORs within each subgroup are reported with 95% CIs. A p for interaction < 0.05 will be described as evidence of effect modification needing confirmation, not as a definite finding.

9. Multiple comparisons

One primary outcome tested at 5%. Secondary outcomes and the subgroup analysis are reported with 95% CIs and interpreted as exploratory; no adjustment for multiplicity.

10. Missing data

Assessment	n (%) missing for every variable and by group; patients with and without lactate compared (SOFA, MAP, vasopressor, ICU, mortality) to judge the mechanism (MCAR vs MAR)
Expected	Lactate missing in 10-20% (not measured in less severe patients and in the ICU without a point-of-care analyser); all other variables < 2%

Main approach	Q1-Q3, Q5: complete case. Q4 (where lactate is the exposure): multiple imputation by chained equations (`mice`), m = 20, predictive mean matching for continuous and logistic regression for binary variables. The imputation model includes the outcome (`died_28d`), all model covariates and the predictors of missingness (SOFA, APACHE II, MAP, vasopressor, ventilation, ICU, source of infection, fluid-balance group). Results pooled with Rubin's rules
Sensitivity	Q1 primary model repeated with multiple imputation; Q4 reported with both complete case and multiple imputation

11. Sample size

FLUID-ICU enrols all eligible patients during 2 years in 4 ICUs (expected about 600). Assumptions: 28-day mortality about 20% in patients with a negative balance, and about 40% of patients have a negative balance (Paper 2 and local audit). With 240 vs 360 patients and two-sided 5% significance, the study has about 80% power to detect an OR of 1.7 (20% vs 30% mortality) and about 55% power for an OR of 1.5. The study is therefore adequately powered for a moderate association only; the 95% CI will show the precision achieved.

12. Software

R version 4.6 with packages tidyverse, gtsummary, flextable, broom, survival, survminer, mice and janitor. Code in `Course/Data/R/` (scripts 01-06), run in order; data cleaning and analysis are fully scripted and reproducible (fixed random seeds).

13. Table and figure shells

Table 1. Baseline characteristics by cumulative fluid balance at day 3

Characteristic	Overall (n = __)	Negative balance (n = __)	Positive balance (n = __)
Age, years, mean (SD)			
Male sex, n (%)			
Heart failure / CKD / diabetes, n (%)			
SOFA score, median (IQR)			
Lactate, mmol/L, median (IQR); missing n			
Vasopressor / mechanical ventilation, n (%)			

Table 2. Outcomes by group and adjusted associations

Outcome	Negative	Positive	Crude effect (95% CI)	Adjusted effect (95% CI)	p
28-day mortality, n (%)			OR __	OR __	

Outcome	Negative	Positive	Crude effect (95% CI)	Adjusted effect (95% CI)	p
AKI (KDIGO >= 1), n (%)			OR per 10 ml/kg __	__	
Ventilator-free days, median (IQR)			Difference __	__	
ICU stay in survivors, days, median (IQR)			Ratio __	__	

Figures

- Figure 1: STROBE participant flow diagram.
- Figure 2: Kaplan-Meier survival to day 28 by fluid-balance group, with numbers at risk and log-rank p.
- Figure 3: Forest plot of adjusted odds ratios (primary model).
- Figure 4: OR for positive balance by heart-failure subgroup, with p for interaction.
- Supplementary: missing lactate by SOFA; complete case vs multiple imputation.

14. Version history and sign-off before data lock

Version	Date	Changes and reason
1.0	Before database lock	First version

Role	Name	Signature	Date
Statistician	[simulated]		
Principal investigator	[simulated]		

Any change after data lock gets a new version, the date and the reason, and is reported in the paper.